



LESSON 5-1

 n th Roots, Radicals, and Rational Exponents

Lesson Overview

Objective

Students will be able to:

- ✓ Use properties of exponents to rewrite expressions involving radicals in terms of rational exponents.
- ✓ Find all real n th roots of a number.
- ✓ Evaluate expressions with rational exponents.
- ✓ Use n th roots to solve equations by rewriting expressions using the properties of exponents.

Essential Understanding

A rational exponent has an equivalent radical expression and any radical expression may be written with a rational exponent. This foundation is necessary to simplify expressions and solve equations.

Previously in course, students:

- Used theorems to find the real and complex roots of polynomial equations.

In this lesson, students:

- Find n th roots of a number and use them to rewrite expressions and solve equations involving rational exponents.

Later in this topic, students will:

- Use the properties of exponents and radicals to simplify radical expressions.

This lesson emphasizes a blend of *conceptual understanding* and *procedural skill and fluency*.

- Students understand that the properties of integer exponents can be extended to rational exponents.
- Students find the n th root of each side of an equation by raising each expression to the $\frac{1}{n}$ power to solve an equation.



Glossary is available at SavvasRealize.com.

Vocabulary Builder

REVIEW VOCABULARY

English | Spanish

- **complex conjugate** | *conjugados complejos*
- **exponent** | *exponente*
- **index** | *índice*
- **radical symbol** | *símbolo radical*
- **radicand** | *radicando*

NEW VOCABULARY

- **n th root** | *n th raíz*

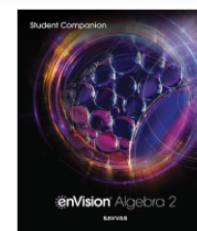
VOCABULARY ACTIVITY

Have students recall their use of n when referring to the n th term of a pattern. Explain that n th roots are denoted by $\sqrt[n]{c}$, where c is the *radicand*, which is inside the *radical symbol*, and n is the *index*. For each of the n th roots shown below, have students identify the *radicand* and the *index*.

$\sqrt[4]{14}$	radicand _____ [14]	index _____ [6]
$\sqrt[3]{27^2}$	radicand _____ [27 ²]	index _____ [3]

Student Companion

Students can do their work for the lesson in their *Student Companion* or on Savvas Realize.



Mathematics Overview

Students relate radical expressions and expressions with rational exponents. They use the properties of exponents to rewrite radical expressions as expressions with a fractional exponent, vice versa.

Students evaluate radical expressions, in either the radical or exponential form, and find all real n th roots of a number. Then students solve equations using n th roots.

Applying Math Practices

Make Sense of Problems and Persevere in Solving Them

Students plan solution pathways through problems when they use n th roots to solve equations involving exponents.

Use Appropriate Tools Strategically

Students recognize the strengths of available tools when they use a calculator to compute rational roots.



Activity

EXPLORE & REASON

INSTRUCTIONAL FOCUS Students explore using a graph to find approximations of square roots and cube roots. This prepares them to evaluate expressions with radicals and rational exponents.

STUDENT COMPANION Students can complete the *Explore & Reason* activity in their *Student Companion*.

Before **WHOLE CLASS**

Implement Tasks that Promote Reasoning and Problem Solving **ETP**

Q: What do you notice about the graph of $y = x^2$?

[The graph is a parabola. All values of y are greater than or equal to zero.]

During **SMALL GROUP**

Support Productive Struggle in Learning Mathematics **ETP**

Q: What is true about the possible missing x -values compared to the possible missing y -values?

[For every ordered pair that is missing an x -value, there are 2 possible values for x . For every ordered pair that is missing a y -value, there is only 1 possible value for y .]

For Early Finishers

Q: Draw a graph of $y = x^4$ and $y = x^5$. Use the graphs to find approximate values as in Part C. How do the numbers of x - and y -values compare to the numbers of x - and y -values of the graphs of $y = x^2$ and $y = x^3$?

[Each y -value for $y = x^4$ has 2 possible x -values, like $y = x^2$. Each y -value for $y = x^5$ has 1 possible x -value, like $y = x^3$.]

After **WHOLE CLASS**

Facilitate Meaningful Mathematical Discourse **ETP**

Q: Can the instructions you created in Part B be used to approximate the cube root of a number?

[Yes; the instructions also work for finding cube roots. Use the graph to find a point with a y -value that is placed under the radical sign. Then find the cube root to find the corresponding x -value of that point.]

Q: What do you notice about the answers in Part C?

[The value for $\sqrt[3]{5}$ is the same as the solution to $x^3 = 5$, the value for $\sqrt[3]{-5}$ is the same as the solution to $x^3 = -5$, and the value for $\sqrt[3]{8}$ is the same as the solution to $x^3 = 8$.]

HABITS OF MIND

Look for Relationships How is $\sqrt{5}$ related to 5? How is $\sqrt[3]{5}$ related to 5? How do you think $\sqrt[4]{5}$ is related to 5?

$[\sqrt[3]{5^2} = 5; \sqrt[3]{5^3} = 5; \sqrt[3]{5^4} = 5]$

Use with EXPLORE & REASON



Explore & Reason is available at SavvasRealize.com.

EXPLORE & REASON

The graph shows $y = x^2$.

A. Find all possible values of x or y so that the point is on the graph.

(a) (2,) (b) (3,)
 (c) (-3,) (d) (5,)
 (e) (, 4) (f) (, -16)
 (g) (, 7) (h) (, 5)

Enter your answer:

B. **Communicate Precisely** Write a precise set of instructions that show how to find an approximate value of $\sqrt{13}$ using the graph.

Enter your answer:

STUDENT EDITION

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nth Roots, Radicals, and Rational Exponents

EXPLORE & REASON

The graph shows $y = x^2$.

A. Find all possible values of x or y so that the point is on the graph.

(a) (2,) (b) (3,)
 (c) (-3,) (d) (5,)
 (e) (, 4) (f) (, -16)
 (g) (, 7) (h) (, 5)

VOCABULARY
 • n th root

B. **Communicate Precisely** Write a precise set of instructions that show how to find an approximate value of $\sqrt{13}$ using the graph.

C. Draw a graph of $y = x^3$. Use the graph to approximate each value.

(a) $\sqrt[3]{5}$ (b) $\sqrt[3]{-5}$
 (c) $\sqrt[3]{8}$ (d) A solution to $x^3 = 5$
 (e) A solution to $x^3 = -5$ (f) A solution to $x^3 = 8$

ESSENTIAL QUESTION How are exponents and radicals used to represent roots of real numbers?

SAMPLE STUDENT WORK

- A. a. 4 b. 9
 c. 9 d. 25
 e. -2, 2 f. no possible value
 g. $-\sqrt{7}, \sqrt{7}$ h. $-\sqrt{5}, \sqrt{5}$

B. Draw the line $y = 13$ on the graph. Draw a vertical line from where the horizontal line intersects the graph down to the positive x -axis, and read the corresponding value.

- C. a. ≈ 1.7
 b. ≈ -1.7
 c. 2
 d. ≈ 1.7
 e. ≈ -1.7
 f. 2



INTRODUCE THE ESSENTIAL QUESTION

Establish Mathematics Goals to Focus Learning ETP

Introduce students to n th roots. Remind them that they have previously found square roots and cube roots. Explain that they can use radicals and exponents to represent and find roots.

EXAMPLE 1 Find All Real n th Roots

Pose Purposeful Questions ETP

Q: When finding the n th root of a number, what do you notice about the real and complex solutions?

[The sum of the number of real and complex solutions is n .]

Q: How are the real roots of an even power related?

[The real roots of an even power are opposites.]

Try It! Answers

1. a. ± 3
b. 4

Elicit and Use Evidence of Student Thinking ETP

Q: What do you know about the roots of a number before you solve the problem?

[If the degree is odd, there will be an odd number of real roots. If the degree is even, there will be an even number of real roots.]

Examples are available at SavvasRealize.com.

ESSENTIAL QUESTION

How are exponential and logarithmic functions related to represent roots of real numbers?

EXAMPLE 1

Find all Real n th Roots

A. What are all the real cube roots of 125?

An **n th root** of a number c is x , such that $x^n = c$. An n th root can be represented using a radical symbol by $\sqrt[n]{c}$.

Since $125 = 5^3$, $\sqrt[3]{125} = 5$ is a real cube root.

To determine if there are other real cube roots of 125, consider the function $f(x) = x^3 - 125$. The zeros of f are the cube roots of 125.

The graph of f has only one x -intercept. Therefore, 5 is the only real cube root of 125.

MAKE SENSE AND PERSEVERE

Recall that third-degree polynomial functions have three solutions. Here, one solution is real and the other two are complex. How could you find the complex solutions?

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Common Error

Try It! 1a When finding the real fourth roots of 81, students may give the root as 3. Have students find the square roots of 81, reminding them that there are two real roots. Then ask students how this is related to finding the real fourth roots of 81.

Additional Examples

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ADDITIONAL EXAMPLE 1

ESSENTIAL QUESTION

Find the real fourth roots of -256.

$$x^4 = -256$$

$$x^4 + 256 = 0$$

$x^4 + 256 = 0$ does not have real solutions.

Therefore, there are no real fourth roots of -256.

SOLUTION

Example 1 Help students find the real cube and fourth roots of a negative number.

Q: Can you find the real cube and fourth roots of a negative number?

[You can find the cube root of a negative number, but there is no real number that when raised to the fourth power results in a negative number.]

Additional Examples are available at SavvasRealize.com.

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ADDITIONAL EXAMPLE 6

ESSENTIAL QUESTION

A clerk is wrapping two cube-shaped packages and wants to know the smallest amount of wrapping paper that can wrap each package. The edge length of the larger package is 3 in. greater than the edge length of the smaller package. The surface area of the larger package is 384 in.^2 . What is the surface area of the smaller package?

$$384 = 6(x + 3)^2$$

$$64 = (x + 3)^2$$

$$8 = x + 3$$

$$5 = x$$

The side length of the smaller cube is 5 in.

$$6(5)^2 = 6(25) = 150$$

so the surface area of the smaller cube is 150 in.^2 .

SOLUTION

Example 6 Help students use n th roots to find the edge length of a cube, given the surface area.

Q: What is the value of n in the n th root in this example? [2]

Q: Why is the expression $6s^2$ used to find the surface area of a cube?

[The area of a face of a cube is equal to s^2 and there are 6 faces on a cube.]



EXAMPLE 2 Understand Rational Exponents

Pose Purposeful Questions ETP

- Q:** Does the order in which you raise a number to a power and find the n th root matter?
[No; you can raise a number to a power and then take the n th root, or you can take the n th root and then raise that number to a power.]
- Q:** Assuming that $\sqrt[n]{x}$ is a real number, what do you know about the value of n if x is negative?
[If x is negative, the value of n is an odd number, because you cannot find real n th roots of a negative number if n is even.]
- Q:** Could a rational exponent have a 1 in the denominator? What does that represent?
[A rational exponent can have a 1 in the denominator. A rational exponent with a 1 in the denominator would be the same as an exponent with an integer value.]

Try It! Answers

2. a. square root of 25; 5
b. fifth root of 32 squared; 4

Elicit and Use Evidence of Student Thinking ETP

- Q:** What do the numerator and denominator of a rational exponent represent?
[The numerator is the power that the number is raised to. The denominator is the root of the number you are finding.]

Use with EXAMPLES 1 & 2

HABITS OF MIND

Generalize If a base raised to a fractional exponent is imaginary, what is true about the values of the numbers involved?

[The base is negative, and the denominator of the fractional exponent must be even.]

Examples are available at SavvasRealize.com.

CONSTRUCT ARGUMENTS
Consider the function $f(x) = x^4 - 16$. How can you use the graph of f to conclude that 16 has two real fourth roots?

CONCEPTUAL UNDERSTANDING

STUDY TIP
It is important to recognize multiple representations of the same value so you can select the form that is most efficient for a given situation.

EXAMPLE 1 CONTINUED

B. What are all the real fourth roots of 16?

Since $16 = 2^4$, one fourth root is the principal root, $\sqrt[4]{16} = 2$.

To determine all the real fourth roots of 16, solve $x^4 = 16$.

$$x^4 = 16$$

$$x^4 - 16 = 0$$

$$(x^2 - 4)(x^2 + 4) = 0$$

$$(x - 2)(x + 2)(x^2 + 4) = 0$$

The zeros of the first two factors are 2 and -2. The zeros of the quadratic factor $x^2 + 4$ are imaginary. Therefore, the real fourth roots of 16 are 2 and -2.

Try It! 1. Find the specified roots of each number.

- a. real fourth roots of 81
- b. real cube roots of 64

EXAMPLE 2 Understand Rational Exponents

A. What is the meaning of the exponent in the expression $16^{\frac{1}{4}}$?

Extend the properties of integer exponents to rational exponents.

$$16^{\frac{1}{4}} = (2^4)^{\frac{1}{4}}$$

$$= 2^1$$

$$= \sqrt[4]{16}$$

$$= 2$$

Therefore, $16^{\frac{1}{4}}$ is another way to represent $\sqrt[4]{16}$, or 2.

In general, for positive integer n , $x^{\frac{1}{n}}$ can be expressed as the radical $\sqrt[n]{x}$.

$$x^{\frac{1}{n}} = \sqrt[n]{x}$$

B. Write $27^{\frac{2}{3}}$ as a radical.

$$27^{\frac{2}{3}} = (27^{\frac{1}{3}})^2$$

$$= (\sqrt[3]{27})^2$$

$$= 3^2$$

$$= 9$$

This means that you can rewrite $27^{\frac{2}{3}}$ as $(\sqrt[3]{27})^2$. Since 27 is a perfect cube, you can evaluate the radical expression mentally by taking the cube root first and then squaring.

$$27^{\frac{2}{3}} = (\sqrt[3]{27})^2 = 3^2 = 9$$

Try It! 2. Explain what each fractional exponent means, then evaluate.

- a. $25^{\frac{1}{2}}$
- b. $32^{\frac{2}{5}}$

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ELL English Language Learners (Use with EXAMPLE 2)

READING BEGINNING Write the following sentences on the board and help students to read them. Laughter is an *expression* of emotion. He sat there with an *expression* of sadness on his face. Although her voice was quiet, her *expression* reflected strength.

- Q:** Using context clues, what does the word *expression* mean in the sentences you read?
[a way of communicating, or representing, emotions without speaking]
- Q:** Have students read the first sentence of both parts (a) and (b). What does the word *expression* mean in the example?
[numbers and symbols that show a value]

LISTENING DEVELOPING Students may think of *power* as the possession of control over others, as illustrated in the following sentence. Read this sentence aloud: The highest *power* in our country is held by the president. Then read the example aloud as students listen.

- Q:** Does *power* mean the same thing in the example as it does in the sentence you heard? [no]
- Q:** What other word did you hear in the example that means the same thing as *power*? [exponent]
- Q:** Listen to the first sentence again. What is the *exponent*? What is the *power*?
[$\frac{1}{4}$, $\frac{1}{4}$. Exponent and power are synonyms.]

WRITING EXPANDING Write the following sentences on the board: Do you own any property? One property of gold is that it is a soft metal.

- Q:** What does the word *property* mean in the first sentence?
[a piece of land that a person owns]
- Q:** What does the word *property* mean in the second sentence?
[a characteristic that describes something]
- Q:** How does the second definition help you understand how the term *property* is used in math?
[Math properties describe the characteristics of numbers and their relationships.]



STEP 2 Understand & Apply

EXAMPLE 3

Evaluate Expressions with Rational Exponents

Pose Purposeful Questions ETP

Q: Why might it be easier to calculate the n th root before raising to a power?

[When you calculate the n th root first, the number that you raise to a power is smaller, so the calculation is easier.]

Q: What is another method you could use to evaluate an expression that has a negative rational exponent?

[You can rewrite the base of the exponent as its reciprocal before or after calculating the root and power.]

Q: How is the Fujita scale classification related to the value of F you calculated?

[The number in the Fujita scale classification is equal to the integer part of the value calculated.]

Q: Why is a calculator a useful tool when approximating a cube root?

[If the cube root is not a perfect cube, it is difficult to accurately approximate the value.]

Try It! Answers

3. a. -8
b. 2.72

Elicit and Use Evidence of Student Thinking ETP

Q: In Part A, does it matter in what order you apply the negative?

[Because the power is within parentheses, you must evaluate the power before applying the negative.]



Examples are available at SavvasRealize.com.

CONCEPT Interpreting Rational Exponents

In general, if the n th root of c is a real number, $\sqrt[n]{c} = c^{\frac{1}{n}}$.

Furthermore, if m is an integer, then

$$c^{\frac{m}{n}} = (\sqrt[n]{c})^m = (\sqrt[n]{c^m}) = (c^m)^{\frac{1}{n}} = c^{\frac{m}{n}}.$$

APPLICATION

USE APPROPRIATE TOOLS

Many expressions with integer roots can be evaluated using mental math. However, when the root is not an integer, it is helpful to use the power function of a calculator to compute rational roots.

EXAMPLE 3 Evaluate Expressions With Rational Exponents

A. Evaluate the expressions $32^{\frac{2}{5}}$, $27^{-\frac{2}{3}}$, and $50^{\frac{1}{4}}$.

$$\begin{aligned} 32^{\frac{2}{5}} &= (32^{\frac{1}{5}})^2 \\ &= 2^2 \\ &= 4 \end{aligned} \quad \begin{aligned} 27^{-\frac{2}{3}} &= (27^{\frac{1}{3}})^{-2} \\ &= (3)^{-2} \\ &= \frac{1}{9} \end{aligned}$$

Since 50 does not have a perfect 4th root, use a calculator to approximate:
 $50^{\frac{1}{4}} \approx 18.80$

B. The Fujita scale rating, F , of a tornado is represented by

$F = \sqrt[3]{\left(\frac{W}{14.1}\right)^2} - 2$, where W is the estimated wind speed of the tornado in miles per hour. What is the Fujita scale rating for this tornado?



$$\begin{aligned} F &= \sqrt[3]{\left(\frac{100}{14.1}\right)^2} - 2 \\ &= \left(\frac{100}{14.1}\right)^{\frac{2}{3}} - 2 \\ &\approx (7.09)^{\frac{2}{3}} - 2 \\ &\approx 3.69 - 2 \\ &\approx 1.69 \end{aligned}$$

A tornado with estimated wind speeds of 100 mph is classified as F1 according to the Fujita scale.

Try It! 3. What is the value of each expression? Round to the nearest hundredth if necessary.

- a. $-(16^{\frac{1}{3}})$ b. $\sqrt[3]{3.5^4}$



Extend Student Thinking

USE WITH EXAMPLE 3 Have students explore finding the value of more complicated expressions that contain rational exponents, expanding their use of rational exponents across the different properties of exponents.

What is the value of each expression?

- $27^{\frac{1}{3}} + 8 - 8^{\frac{2}{3}}$
[85]
- $4^{\frac{1}{2}} + 5 - 16^{\frac{1}{2}}$
[-27]
- $3(8)^{-\frac{1}{3}} + 5 + 4^{-\frac{1}{2}}$
[7]
- $\left(\frac{8}{27}\right)^{-\frac{1}{3}} - 6 - \left(\frac{81}{16}\right)^{\frac{1}{4}}$
[-6]
- $(64^{\frac{1}{3}})^{\frac{1}{2}}$
[32]



EXAMPLE 4 Simplify n th Roots

Pose Purposeful Questions ETP

PART A

Q: What is another approach you can use to simplify the expression? [You can write the 5th root as the rational exponent $\frac{1}{5}$. Then multiply the exponent of each factor by the rational exponent and simplify.]

PART B

Q: Why is it necessary to assume all variables represent positive numbers?

[Because the given expression uses the radical symbol, which represents the principal (positive) root, it can only be equivalent to an expression with a nonnegative value. The exponent of x^5 is odd, so the value of x^5 could be negative. In this case, the expression x^5y^2 would not be equivalent to the given expression because it would be negative. Using variables to represent only positive numbers ensures that the resulting expression is nonnegative and thus equivalent to the given expression.]

Try It! Answers

- $-2ab^3$
- $4x^3y^6$

Elicit and Use Evidence of Student Thinking ETP

Q: Why is it important to write the radicand as an expression to a power before evaluating?

[Rewriting as an expression to a power makes it simpler to find the n th root.]

EXAMPLE 5 Use n th Roots to Solve Equations

Pose Purposeful Questions ETP

Q: What property are you using when you rewrite $(x^5)^{\frac{1}{5}}$ as x ? [the Power of a Power Property]

Try It! Answers

- $x = 4$
- $p = \pm 3$

Elicit and Use Evidence of Student Thinking ETP

Q: What do you notice about these equations that makes them easier to solve?

[The coefficient of the variable divides the value on the other side of the equation evenly. In part (a), this quotient is a perfect cube. In part (b), this quotient is a perfect fourth power.]



Examples are available at SavvasRealize.com.

EXAMPLE 4 Simplify n th Roots

Simplify each expression. Assume all variables are positive.

A. $\sqrt[5]{32m^{15}}$

$$\sqrt[5]{2^5m^{15}} = \sqrt[5]{(2m^3)^5}$$

$$= 2m^3$$

$$\text{So, } \sqrt[5]{32m^{15}} = 2m^3.$$

Write the radicand as an expression to the 5th power.

B. $\sqrt[4]{x^{20}y^8}$

$$\sqrt[4]{(x^5y^2)^4} = (x^5y^2)^1$$

$$= x^5y^2$$

$$\text{So, } \sqrt[4]{x^{20}y^8} = x^5y^2.$$

Raise the radicand to the one-fourth power.

USE STRUCTURE

You can also work this problem using the method in Part A. The exponents 20 and 8 are both multiples of 4. So you can rewrite this expression as $\sqrt[4]{(x^5y^2)^4}$ which equals x^5y^2 .

Try It! 4. Simplify each expression.

a. $\sqrt[3]{-8a^3b^9}$

b. $\sqrt[5]{256x^{12}y^{24}}$

EXAMPLE 5 Use n th Roots to Solve Equations

Solve the equation $2x^5 = 64$.

A. $2x^5 = 64$

$$x^5 = 32$$

$$(x^5)^{\frac{1}{5}} = 32^{\frac{1}{5}}$$

$$x = 2$$

Raise each side of the equation to the $\frac{1}{5}$ power.

The graph shows one point of intersection, so there is only one real solution.

The solution to the equation is $x = 2$.

B. $3x^4 = 1875$

$$x^4 = 625$$

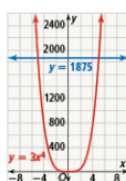
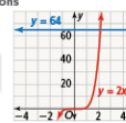
$$(x^4)^{\frac{1}{4}} = \pm(625)^{\frac{1}{4}}$$

$$x = \pm 5$$

A negative integer raised to an even power is positive, so -5 is also a real solution.

The graphs show 2 points of intersection, so there are no other real solutions.

The solutions are $x = \pm 5$.



Try It! 5. a. Solve the equation $5x^3 = 320$.

b. Solve the equation $2p^4 = 162$.

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HABITS OF MIND

Use with EXAMPLES 3 & 4

Make Sense and Persevere What is an example of a variable expression that has both a cube root and a fourth root, which can be simplified to an expression without a radical?

[Answers will vary. Sample: x^{12}]



Support Struggling Students

USE WITH EXAMPLE 4 Students may have difficulty simplifying expressions with n th roots. Have students practice simplifying expressions with square roots.

Simplify each expression. Assume all variables are positive.

1. $\sqrt{4a^4}$
[$2a^2$]

2. $\sqrt{49y^6z^2}$
[$7y^3z$]

3. $\sqrt{121m^8n^4}$
[$11m^4n^2$]



STEP 2 Understand & Apply

EXAMPLE 6 Use n th Roots to Solve Problems

Use and Connect Mathematical Representations **ETP**

Q: What does *volume* mean in the context of this problem?

[The volume is the amount of water that a cube can hold.]

Q: What do you notice about the image that is important to understanding and solving this problem?

[The top cube is larger than the smaller cube. To find the difference of the volumes of the cubes, you have to subtract the volume of the smaller cube from the volume of the larger cube.]

Try It! Answer

6. $\approx 692.69 \text{ cm}^3$

Elicit and Use Evidence of Student Thinking **ETP**

Q: Is the edge length of the smaller cube an integer? How can you tell without solving?

[No; you can tell because the value of the volume, 200 cm^3 , is not a perfect cube.]

Use with EXAMPLES 5 & 6

HABITS OF MIND

Communicate Precisely What are the steps necessary to solve the equation $ax^n = b$?

[Divide both sides of the equation by a to isolate the x -term.

Then take the n th root of both sides so you have $x = \sqrt[n]{\frac{b}{a}}$.]



Examples are available at SavvasRealize.com.

CONCEPT Solving an Equation in the Form $x^n = c$

To solve an equation in the form $x^n = c$, find the n th root of both sides by raising each expression to the $\frac{1}{n}$ power.

$$(x^n)^{\frac{1}{n}} = (c)^{\frac{1}{n}}$$

APPLICATION

EXAMPLE 6 Use n th Roots to Solve Problems

A garden water fountain is made of two cubes. The larger container has an edge length 2 cm longer than the edge length of a second cube. When the larger cube empties into the smaller cube, how much water will spill over the edge of the smaller cube into the base?



MODEL WITH MATHEMATICS

Use the information in the problem to write an equation that models the situation.

$$(x + 2)^3 = 729$$

$$[(x + 2)^3]^{\frac{1}{3}} = (729)^{\frac{1}{3}}$$

$$x + 2 = 9$$

$$x = 7$$

Raise both sides to the $\frac{1}{3}$ power and solve.

The volume of the smaller cube is 7^3 , or 343 cm^3 .

The volume of the larger cube is 729 cm^3 .

The amount of water that spills is $729 - 343 = 386$.

When the larger cube empties into the smaller cube, 386 cm^3 of water will spill out into the base of the fountain.

Try It! 6. One cube has an edge length 3 cm shorter than the edge length of a second cube. The volume of the smaller cube is 200 cm^3 . What is the volume of the larger cube?



CONCEPT SUMMARY n th Roots, Radicals, and Rational Exponents

Q: How can you check that you have rewritten a numerical expression in radical form as an expression in exponential form correctly?

[You can evaluate each expression. If the expressions are equal, you have rewritten them correctly.]

Do You UNDERSTAND? | Do You KNOW HOW?

Common Error

Exercise 6 When writing an expression in radical form, students may confuse which number in the fraction represents the power and which number represents the index of the radical. Have students rewrite the fraction as $4 \cdot \frac{1}{3}$. Then ask them which value represents a power and which value represents a root.

Answers

- A radical can be rewritten as a fractional exponent, where the index is the denominator of the exponent. Both a radical with index n and a fractional exponent $\frac{1}{n}$ represent the n th root of a real number.
- Kaitlyn made the exponent equal to the index. The index should be the denominator of the exponent, so the exponent should be $\frac{1}{3}$.
- The index is 5 and the radicand is 125.
- $(75^{\frac{1}{3}})^3$ can be simplified using the Power of a Power Property. Multiplying the exponents gives 75^1 .
- No; Amani did not apply the Power of a Power Property correctly; $(x^8)^{\frac{1}{2}} = x^{8 \cdot \frac{1}{2}} = x^4$.
- Yes, a rational exponent may be an improper fraction; $27^{\frac{4}{3}}$ is evaluated by finding the cube root of 27, which is 3, and raising it to the fourth power, which is 81.
- $\sqrt[5]{a}$
- $\sqrt[3]{7^2}$
- $b^{\frac{1}{3}}$
- $p^{\frac{1}{4}}$
- one
- two
- $x \approx 4.33$
- $x \approx 5.95$
- $3x^4y^2$
- $-2xy^6$
- 27



Concept Summary, Do You Understand, and Do You Know How are available at SavvasRealize.com.

CONCEPT SUMMARY n th Roots, Radicals, and Rational Exponents

Relating Radical and Exponential Forms		Solving an Equation in the Form $x^n = c$
WORDS	The index of a radical is equivalent to the denominator of a fractional exponent. The exponent of the radicand is equivalent to the numerator of a fractional exponent.	To solve an equation in the form $x^n = c$, find the n th root of both sides of the equation by raising each expression to the $\frac{1}{n}$ power.
NUMBERS	Radical Form $\sqrt[3]{32^4} = (32^4)^{\frac{1}{3}} = 32^{\frac{4}{3}}$ Exponential Form $729^{\frac{5}{3}} = (729^{\frac{1}{3}})^5 = (\sqrt[3]{729})^5$	$x^3 = 1,728$ $(x^3)^{\frac{1}{3}} = (1,728)^{\frac{1}{3}}$ $x = 12$
ALGEBRA	Radical Form $\sqrt[n]{c^m} = (c^m)^{\frac{1}{n}} = c^{\frac{m}{n}}$ Exponential Form $c^{\frac{m}{n}} = (c^{\frac{1}{n}})^m = \sqrt[n]{c^m}$	$x^n = c$ $(x^n)^{\frac{1}{n}} = (c)^{\frac{1}{n}}$ $x = c^{\frac{1}{n}}$

Do You UNDERSTAND?

- ESSENTIAL QUESTION** How are exponents and radicals used to represent roots of real numbers?
- Error Analysis** Kaitlyn said $\sqrt[3]{10} = 10^{\frac{1}{3}}$. Explain Kaitlyn's error.
- Vocabulary** In the radical expression $\sqrt[3]{125}$, what is the index? What is the radicand?
- Use Structure** Why is $75^{\frac{1}{3}}$ equal to $(75^{\frac{1}{3}})^3$?
- Construct Arguments** Amani said that $(x^8)^{\frac{1}{2}} = \frac{x^8}{2} = x^4$. Is Amani correct? Explain.
- Make Sense and Persevere** Is it possible for a rational exponent to be an improper fraction? Explain how $27^{\frac{4}{3}}$ is evaluated or why it cannot be evaluated.

Do You KNOW HOW?

Write each expression in radical form.

7. $a^{\frac{1}{2}}$ 8. $7^{\frac{2}{3}}$

Write each expression in exponential form.

9. $\sqrt{b}$ 10. $\sqrt[3]{p^7}$

11. How many real third roots does 1,728 have?

12. How many real sixth roots does 15,625 have?

13. Solve the equation $4x^3 = 324$.

14. Solve the equation $2x^4 = 2,500$.

Simplify each expression. Assume all variables are positive.

15. $\sqrt[3]{27x^{12}y^6}$ 16. $\sqrt[4]{-32x^8y^{30}}$

17. A snow globe is packaged in a cubic container that has volume 64 in.³ A large shipping container is also a cube, and its edge length is 8 inches longer than the edge length of the snow globe container. How many snow globes can fit into the larger shipping container?



STEP 3 Practice & Problem Solving

PRACTICE & PROBLEM SOLVING

Lesson Practice You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: **Lesson Practice**

Adaptive Practice

You may also take advantage of the bank of exercises for assigning additional practice.

Assignment Guide

On-Level	Advanced
18, 20, 22, 25–50	18–50

Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways.

For example:

Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.

Understand, Apply 2 points each

Practice 1 point each

Assessment Practice 1 point each

Performance Task 3 points

Item Analysis

Example	Items	DOK
1	25–28	1
	22	2
2	29–32, 48	1
	20, 21, 24, 47	2
3	33–34	1
4	35–38, 49	1
	18	2
5	39–42	1
6	19, 23, 43–46	2
	50	3

Answers

18. Yes, Justice is correct;
 $(3x^3y)^5 = 3^5(x^3)^5y^5 = 243x^{15}y^5$;



Practice & Problem Solving and video tutorials are available at SavvasRealize.com.

PRACTICE & PROBLEM SOLVING

UNDERSTAND

18. **Construct Arguments** Justice found that the fifth root of $243x^{15}y^5$ is $3x^3y$. Is Justice correct? Explain your reasoning.
19. **Make Sense and Persevere** For a show, spheres were bounced and passed. Explain how to find the radius r of one of the inflated spheres. Use technology to compute your answer.



20. **Error Analysis** Describe and correct the error a student made in writing this exponential expression in radical form.

$$\begin{aligned} x^{\frac{4}{3}} &= (x^4)^{\frac{1}{3}} \\ (x^4)^{\frac{1}{3}} &= \sqrt[3]{x^4} \end{aligned}$$

21. **Construct Arguments** Determine whether $\sqrt[3]{x^2}$ is equal to $(\sqrt[3]{x})^2$. Explain your reasoning.
22. **Use Structure** How many third roots does -512 have? Explain your reasoning.
23. **Higher Order Thinking** The annual interest formula below calculates the final balance of an account, F , given a starting balance, S , and an interest rate, r , after 10 years.

$$F = S(1 + r)^{10}$$

When solving for r , why can the negative root be ignored?

24. **Mathematical Connections** The lengths of the two legs of a right triangle are 4 and 8. What is the length of the hypotenuse, in simplest radical form?

PRACTICE

Find the specified roots of each number.

SEE EXAMPLE 1

25. the real fourth roots of 81 ± 3
26. the real third roots of 343 ± 7
27. the real fifth roots of $1,024 \pm 4$
28. the real square roots of 25 ± 5

Rewrite each expression using a fractional exponent. SEE EXAMPLE 2

29. $\sqrt[4]{16^2}$ $16^{\frac{2}{4}} = 16^{\frac{1}{2}}$ 30. $\sqrt[3]{729}$ $729^{\frac{1}{3}}$
31. $\sqrt[4]{x^2}$ $x^{\frac{2}{4}}$ 32. $\sqrt[4]{ab}$ $(ab)^{\frac{1}{4}}$

What is the value of each expression? Round to the nearest hundredth if necessary. SEE EXAMPLE 3

33. $\sqrt[4]{25^2}$ 5 34. $-\sqrt[3]{125^5}$ $-3,125$

Simplify each expression. Assume all variables are positive. SEE EXAMPLE 4

35. $\sqrt[3]{8y^9}$ $2y^3$ 36. $\sqrt[4]{q^{12}z^4}$ q^3z
37. $\sqrt[4]{729a^{24}b^{18}}$ $3a^6b^3$ 38. $\sqrt[5]{v^8g^{40}}$ vg^8

Solve each equation. SEE EXAMPLE 5

39. $1,125 = 9x^3$ 5 40. $6,480 = 5w^4$ ± 6
41. $270 = 10q^3$ 3 42. $256 = 4h^6$ ± 2

43. A small cube holds chocolate candies. Its side length is 1.5 in. smaller than a second, larger cube of chocolate candies. What is the volume of the larger cube? SEE EXAMPLE 6 $\approx 205 \text{ in.}^3$

Volume is 85 in.³



LESSON 5-1 n th Roots, Radicals, and Rational Exponents 233

19. Use the volume formula to solve for r . Divide $4,186\frac{2}{3}$ by $\frac{4}{3}$ and the result by $\pi \approx 3.14$. This leaves $1,000 \approx r^3$. The cube root of 1,000 is 10, so the radius is approximately 10 in.
20. $(x^4)^{\frac{1}{3}}$ is correct, but the denominator of the fraction, 3, is the index of the radical: $\sqrt[3]{x^4}$.
21. Yes, they are equal; $\sqrt[3]{x^2}$ and $(\sqrt[3]{x})^2$ can both be written as $x^{\frac{2}{3}}$, so they must be equal.
22. 1; only one real number -8 times itself 3 times equals -512 .
23. The negative root can be ignored because the interest rate must be a positive number.
24. $4\sqrt{5}$



Practice

Answers

44. 1 m
 45. 12 in.
 46. 0.025 or 2.5%
 47. $\left(\frac{HW}{3,600}\right)^{\frac{1}{2}}$
 50. Part A 36π ft²
 Part B 3 containers

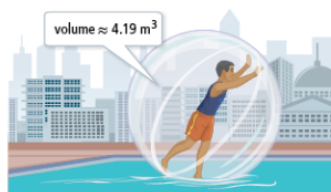


Practice & Problem Solving is available at SavvasRealize.com.

PRACTICE & PROBLEM SOLVING

APPLY

44. **Model With Mathematics** A water-walking ball has a set volume. What is the radius, r , of the ball?



45. **Make Sense and Persevere** Ahmed received a box of gifts. The box is a rectangular prism with the same height and width, and the length is twice the width. The volume of the box is 3,456 in.³. What is the height of the box?



46. **Make Sense and Persevere** Amelia's bank account earns interest annually. The equation shows her starting balance of \$200 and her balance at the end of four years, \$220.82. At what rate, r , did Amelia earn interest?

$$220.82 = 200(1 + r)^4$$

47. **Model With Mathematics** One measure of a patient's body surface area is found using the expression $\sqrt{\frac{H \cdot W}{3,600}}$. Write this with a fractional exponent.

ASSESSMENT PRACTICE

48. Determine if each expression is another way to write $b^{\frac{3}{4}}$. Select Yes or No.

	Yes	No
a. $\sqrt[4]{b^3}$	☒	☐
b. $(b^3)^{\frac{1}{4}}$	☐	☒
c. $b^{\frac{3}{4}}$	☒	☐
d. $\sqrt[3]{b^4}$	☐	☒
e. $\frac{b^3}{b^4}$	☐	☒

49. **SAT/ACT** Which of the following is equivalent to $\sqrt[6]{4,096x^{18}y^{30}}$, where $x > 0$ and $y > 0$?

- Ⓐ $682.7x^{15}y^{24}$
 Ⓑ $4x^{1.6}y^{1.8}$
 Ⓒ $4,096x^3y^5$
 Ⓓ $4x^3y^5$
 Ⓔ $682.7x^3y^5$

50. **Performance Task** A milk processing company uses cylindrical-shaped containers. The height of the container is equal to the diameter of the base.



Part A How much material is needed to make the lateral surface area of the shipping container?

Part B The cargo hold of a ship is 20 ft high. What is the largest number of these shipping containers that could be stacked on top of each other inside the cargo hold?



STEP 4 Assess & Differentiate

LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.

Students can take the Lesson Quiz online or you can download a printable copy from [SavvasRealize.com](https://www.savvasrealize.com). The Lesson Quiz is also available in the *Assessment Sourcebook*.

Item Analysis

Item	DOK	Skills Review and Practice
1	1	A83
2	2	A82
3	2	A82
4	1	A80
5	2	A84



Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

I Intervention	0–3 points	• Reteach to Build Understanding • Mathematical Literacy and Vocabulary • Lesson Virtual Nerd videos
O On-Level	4 points	• Enrichment
A Advanced	5 points	• Enrichment



Lesson Quiz is available at [SavvasRealize.com](https://www.savvasrealize.com).

Name _____

enVision Algebra 2

SavvasRealize.com

5-1 Lesson Quiz

n th Roots, Radicals, and Rational Exponents

1. What are the real fourth roots of 256?

Real fourth roots and

2. Select all expressions that are equivalent to $x^{\frac{1}{2}}$.

☒ A. $(x^3)^{\frac{1}{4}}$

☐ B. $x^{\frac{8}{6}}$

☐ C. $(x^{\frac{1}{2}})^{\frac{2}{4}}$

☒ D. $(x^{\frac{1}{4}})^3$

☐ E. $\sqrt[3]{x^4}$

☒ F. $\sqrt[4]{x^3}$

3. Explain what $243^{\frac{1}{5}}$ means, then evaluate it.

$243^{\frac{1}{5}}$ is the
☐ square
☐ cube
☐ fourth
☒ fifth

root of 243
☐ squared
☒ cubed
☐ to the fourth power
☐ to the fifth power

Its value is
☐ 3
☐ 9
☒ 27
☐ 81

4. Simplify the expression $\sqrt[4]{81m^{12}n^4}$. Assume all variables are positive.

☒ A. $9m^3n$

☐ B. $9m^6n^2$

☒ C. $3m^3n$

☐ D. $3m^8$

5. Solve the equation $-4x^3 = 32$.

$x =$

enVision® Algebra 2 • Assessment Sourcebook



DIFFERENTIATED RESOURCES

I = Intervention **O** = On-Level **A** = Advanced

Reteach to Build Understanding **I**

Provides scaffolded reteaching for the key lesson concepts.

MathXL® for School

Name _____

5-1 Reteach to Build Understanding
nth Roots, Radicals, and Rational Exponents

1. Use the pattern in the top row to convert from the radical form to the exponential form. Fill in the blanks in the table to complete the table.

Radical	Exponential	Rational Exponent
$\sqrt[3]{27}$	$27^{1/3}$	3^3
$\sqrt[4]{16}$	$16^{1/4}$	2^4
$\sqrt[5]{32}$	$32^{1/5}$	2^5

Remember that the index of a radical, n , is equivalent to the denominator of a fractional exponent. The exponent of the radicand, m , is equivalent to the numerator of a fractional exponent.

2. Charlie needed to simplify $(64x^3y^{12})^{1/3}$. First, he found the square root of the term $8x^3y^6$. Then he cubed each term, which gave him a final answer of $512x^9y^{18}$. What mistake did he make and what should he have done instead?

Sample: Charlie should have found the cube root of each term and then squared each one. (Student may also state that Charlie should have squared and then found the cube root. Both would be correct.)

3. Choose the correct answer from the following options, if Charlie had not made a mistake.

a. What mistake did he make and what should he have done instead?

b. Choose the correct answer from the following options, if Charlie had not made a mistake.

Step 1: $5x^4 - 405 = 0$
Step 2: $5x^4 - 405 = 0$
Step 3: $5(x^4 - 81) = 0$
Step 4: $(x^4 - 81) = 0$
Step 5: $(x^2 + 9)(x^2 - 9) = 0$
Step 6: $(x^2 + 9)(x - 3)(x + 3) = 0$
Step 7: $x = \pm 3$

Write the original equation.
Rearrange to equal zero.
Factor out 5.
Divide both sides by 5.
Factor using the conjugate identity.
Factor the right parentheses using the difference of squares pattern.
Set each factor equal to zero and solve.

MathXL® Algebra 2 • Teaching Resources

Additional Practice **I O**

Provides extra practice for each lesson.

MathXL® for School

Name _____

5-1 Additional Practice
nth Roots, Radicals, and Rational Exponents

Find the specified roots of each number.

1. real fourth roots of 625
-5 and 5

2. real cube roots of 125
5

Explain what the fractional exponent means, then evaluate.

3. $144^{1/2}$
12

4. $121^{1/2}$
11

Take square root of 144; 12. Then raise the result to 3; 1,331

What are the values of each expression? Round to the nearest hundredth.

5. $-(64)^{1/3}$
-4

6. $(16)^{3/4}$
8

Simplify each expression. Assume all variables are positive.

7. $\sqrt[3]{-27m^3n^3}$
 $-3mn$

8. $\sqrt[3]{625x^{15}y^{24}}$
 $5x^5y^8$

9. $\sqrt[3]{64x^3y^{18}}$
 $2x^1y^6$

Solve the equations.

10. $7x^3 = 189$
 $x = 3$

11. $199,927 = 7x^4$
 $x = \pm 13$

12. One cube has an edge length 5 cm shorter than the edge length of the second cube. The volume of the smaller cube is 216 cm³. What is the volume of the larger cube?
1,331 cm³

13. Describe and correct the error a student made in writing this expression in radical form.
 $x^4 = (x^2)^2$
The 4 and 5 in the last radical should be interchanged.

14. A water-walking ball has a volume of approximately 984.32 ft³. What is the radius of the ball?
 $(V = \frac{4}{3}\pi r^3)$ 6 feet

15. Jeanne's bank account earns interest annually. The equation below shows her starting balance of \$400 and her balance at the end of five years, \$535.29. At what rate r did Jeanne earn interest?
 $535.29 = 400(1 + r)^5$ **$r = 6\%$**

MathXL® Algebra 2 • Teaching Resources

Enrichment **O A**

Presents engaging problems and activities that extend the lesson concepts.

MathXL® for School

Name _____

5-1 Enrichment
nth Roots, Radicals, and Rational Exponents

The following form of radicals is called a **nested radical**. The number of radicals may extend to infinity.

Example: $\sqrt{5 + \sqrt{5 + \sqrt{5 + \dots}}}$

1. Which equation has a solution equal to $A = \sqrt{1 + \sqrt{1}}$?
a. $(A - 1)^2 = 1$ b. $(A^2 - 1)^2 = 1$ c. $(A^2 + 1)^2 = 1$

2. Which equation has a solution equal to $B = \sqrt{1 + \sqrt{1 + \sqrt{1}}}$?
a. $(B^4 - 2B^2)^2 = 1$ b. $(B^4 - 2B)^2 = 1$ c. $(B^4 - B^2)^2 = 1$

3. Let $C = \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}}$. What is the equivalent of $\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$ in terms of C ?
 $C^2 - 4C^2 + 2$

4. Let $X = \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}}$, where the radical is repeated infinitely many times. Squaring both sides results in $X^2 = 2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$. How does removing one of the radicals affect the number of the radicals remaining? **It does not affect the number of radicals, since there are still infinitely many radicals.**

5. The solution of which equation is equal to the value of X in Problem 4?
a. $X^4 - X^2 - 2 = 0$ b. $X^4 - X^2 - 2 = 0$ c. $X^2 - X - 2 = 0$

MathXL® Algebra 2 • Teaching Resources

Mathematical Literacy and Vocabulary **I O**

Helps students develop and reinforce understanding of key terms and concepts.

Name _____

5-1 Mathematical Literacy and Vocabulary
Properties of Exponents and Radicals

Complete the vocabulary chart by filling in the missing information.

Word or Word Phrase	Definition	Picture or Example
index	The number that gives the degree of the root. It is equivalent to the denominator of a fractional exponent.	1. Answers may vary. Sample: $\sqrt[5]{125}$; 5 is the index of the given radical expression.
nth root	Given the equation $a^n = b$, a is the n th root of b .	2. Answers may vary. Sample: $\sqrt[4]{81} = 3$; the principal fourth root of 81 is equal to 3.
radicand	The value written under the radical symbol.	3. The value written under the radical symbol is 64.
radical symbol	A symbol used to indicate that a root is being taken.	4. Answers may vary. Sample: $\sqrt[3]{125}$; the sign $\sqrt[3]{}$ indicates that the third root of 125 must be taken.
real root	Negative numbers do not have real roots of even indices.	5. Answers may vary. Sample: $\sqrt[6]{-64}$; The sixth root (even index) of a negative number (-64) is not a real root.

MathXL® Algebra 2 • Teaching Resources

Digital Differentiated Resources

The Reteach to Build Understanding, Additional Practice, and Enrichment activities are available as digital assignments. These activities are automatically assigned when students complete the lesson quiz online and are automatically scored.

